

Modern Geometry II course notes

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“It is not knowledge, but the act of learning, not possession but the act of getting there, which grants the greatest enjoyment.”

—Karl Friedrich Gauss, 1808

Acknowledgements

The materials used in these course notes have been drawn from a variety of sources. The notes are meant to be accompanied by the book *Experiencing Geometry In Euclidean, Spherical, and Hyperbolic Spaces* [Henderson & Tamiņa, 2005] and they incorporate many ideas from this book. Several of the problems from this book are reprinted in Appendix C. Section 4 explores the non-Euclid java application created by Joel Castellanos, Joe Dan Austin, and Ervan Darnell. When working on Problems 8.3 and 8.4, I encourage students to use Spherical Easel, a java application created by David Austin and William Dickinson. As discussed in Section 6, we usually also spend some time exploring the program KaleidoTile, which was created by Jeff Weeks. The activity in Section 7 is a slightly modified version of Project 64 from the booklet *101 Project Ideas for The Geometers Sketchpad* [?]. The assignment in Section 9 discusses an excerpt from *Gödel, Escher, Bach: an Eternal Golden Braid* [Hofstadter, 1999] which is reprinted in Appendix D.

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Preface

“The royal road to knowledge,
it is easy to express:
to err, and err, and err again,
but less, and less, and less.”

This is the second course in a two course sequence that focuses on geometry and learning to develop and express your own mathematical ideas. Like the first course, it will emphasize ideas and imagination in addition to techniques and calculations. We will be further investigating Euclidean and Spherical Geometry, as well as other geometries, such as Taxicab Geometry and Hyperbolic Geometry. Like the first semester course, this course will focus on mathematical reasoning and communication skills in addition to mathematical facts.

These notes are meant to be accompanied by the textbook *Experiencing Geometry: Euclidean and Non-Euclidean with History* by David W. Henderson and Daina Taimina [Henderson & Taimina, 2005]. A lot of this book may already look familiar to you if you already took a course using the first part of these notes. Normally, the first semester of this course covers much of the material in chapters 1–4, 6, and 7. This course will look at significant parts of chapters 5, 8, and 10, and some other sections.

Chapter 1

Formal Assignment 1: Taxicab Geometry

Write a group paper about taxicab geometry. In your paper, you should explain what Taxicab geometry is and what you have learned about it. You can look at some of the questions below, or any other questions that seem interesting to you.

Some possible questions:

- Can you give a formula for taxicab distance? Is taxicab distance always less or more than Euclidean distance? Can you prove your answer?
- What do taxicab circles look like? How can they be used? What is the taxicab value of π ?
- What do taxicab ellipses look like? How can they be used? (Recall, an ellipse with foci A and B is $\{P \mid d(A, P) + d(B, P) = c\}$, the set of all of the points that can be drawn using a string of length c attached to points A and B .)
- What happens if we look instead at a set $\{P \mid d(A, P) - d(B, P) = c\}$ in Euclidean or taxicab geometry? What happens in the case where $c = 0$?
- How do you find the distance from a point to a line in taxicab geometry? Can you give a careful procedure that always works? How does this differ from euclidean geometry? If I have a point A and a line L , what is $\{P \mid d(A, P) = d(P, L)\}$ in Euclidean or taxicab geometry?
- Can you use Geometer's Sketchpad to model any of these shapes in taxicab geometry?
- Are there any theorems of Euclidean geometry that aren't true in taxicab geometry?

Chapter 2

Informal Assignment 1: Build your own Hyperbolic Plane!

Build your own hyperbolic Plane! You can use any one of the four methods explained in appendix B of *Experiencing Geometry*. The crocheted hyperbolic planes are by far the nicest, but the hardest to make. This assignment will be graded on the quality of your finished product.

Chapter 3

Informal Assignment 2: Straightness and Euclid's Postulates on the Hyperbolic Plane

Decide which lines are straight on your Hyperbolic Plane. In doing so, you may find it helpful to look at Problem 5.1. Which of Euclid's five Postulates are true on the Hyperbolic Plane? Also look at the modified version of his first postulate that we will call Postulate 1': Two points determine a unique line.

Your write-up of this assignment should include explanations of which lines are straight on the hyperbolic plane and how you know they are straight, and for each postulate, whether it is true or false, and how you know.

Chapter 4

Informal Assignment 3: Non-Euclid

Go to the webpage for the non-Euclid applet, which is found at <http://www.cs.unm.edu/~joel/NonEuclid/NonEuclid.html>.

This applet provides a dynamic geometry model of one kind of hyperbolic geometry, called the Poincaré Disk model. Notice that it is a kind of taxicab geometry: the distance between two points in the Poincaré disk is given by a non-Euclidean distance function.

Experiment with this model. It seems very different from the hyperbolic planes that we have been looking at. Do you think it is reasonable to call this a model of hyperbolic geometry? Why or why not? You may also find it useful to compare this model with the other model provided by non-Euclid, which is called the Upper Half-Plane model.

Try to come up with as many examples as you can of theorems of Euclidean geometry that are not true in the Poincaré Disk model, and of facts that appear to be true in the Poincaré Disk model that are not true in Euclidean geometry.

Chapter 5

Informal Assignment 4: Folding a parallel line

5.1 Part 1

Assume that you have been given a piece of paper containing a straight line and a point not on that straight line. Figure out how to create a straight line through the point that is parallel to the original line, using only paper folding. Make sure that your method doesn't require any estimation on the part of the person doing the folding. Write up your method in a series of steps that someone else could follow.

5.2 Part 2

Prove that your method always works—that is, that the line created by your method can never intersect the original line. Your argument should not include any facts that you don't know how to prove. You will probably find Problem 8.2 helpful here.

Chapter 6

Regular Tessellations and Polyhedra

You have been given a set of polydrons (snap-together polygons) to work with. We are going to use the polydrons to produce **regular** figures—that is, figures where the same number of a given regular polygon meet at each vertex. Using your polydrons, complete the following table. If a given combination produces a planar tessellation, draw it; if it produces a polyhedron, name it according to how many faces it has.

# of shapes meeting at each vertex	3	4	5	6	7
Triangles					
Squares					
Pentagons					
Hexagons					
Heptagons					

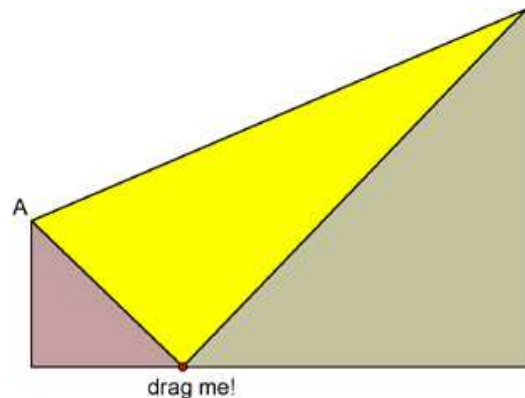
The following list of greek numerical prefixes may be useful in naming the polyhedra:

1	mono
2	di
3	tri
4	tetra
5	penta
6	hexa
7	hepta
8	octa
9	ennea
10	deca
11	hendeca
12	dodeca
13	triskaideca
14	tetrakaideca
15	pentakaideca
16	hexakaideca
17	heptakaideca
18	octakaideca
19	enneakaideca
20	icosa
24	icositetra
30	triconta or triaconta
40	tetraconta
50	pentaconta

Chapter 7

Computer Lab I: Paper folding

When the top left corner of a piece of paper is folded down to touch the bottom edge of the paper, a triangle is formed in the lower left corner, as shown below. Model this situation in Sketchpad. Now use the sketch to answer the following question: When does the lower-left triangle have the biggest possible area?



Further challenges:

1. Can you prove that your answer is correct? It may be helpful to recall that you probably learned about maximization problems in a calculus class.
2. Your sketch probably falls apart when the point is dragged too close to the lower-right corner. Extend your sketch to show (and smoothly transition into) this third possible state. Can you construct a sketch in which the corner point can be dragged all the way around the edge of the piece of paper?

What to turn in:

- Turn in a sketchpad file electronically by emailing it to your instructor. See how pretty you can make it! The file you submit should be named by your last name(s).
- Also turn in a written explanation of when the triangle is as big as possible, and your proof that you're right.

Chapter 8

Computer Lab II: The Product of Two Reflections

Examine what happens when you reflect a figure consecutively over two separate lines. What is the resulting motion of the figure? State your findings as precisely as possible, and see if you can prove them. Be sure to account for all possible arrangements of the lines.

Chapter 9

Informal Assignment 9: Parallel Postulates

An excerpt from the book *Gödel, Escher, Bach: an Eternal Golden Braid* [Hofstadter, 1999] is reproduced in Appendix D of these notes. On the last page of this excerpt, Douglas Hofstadter claims, without explanation, that Euclid's Fifth Postulate (EFP) and the Parallel Postulate (PP) are "equivalent." However, he doesn't say what he means by the word "equivalent" in this context. Show that this definition matters by giving a careful definition of the word "equivalent" under which these statements are not equivalent, and explain why; then give a (different) careful definition of this word under which these statements are equivalent, and prove that they are in fact equivalent under your definition.

In answering this question, you may find it helpful to read *Experiencing Geometry* Problem 10.3; the assumption that Hofstadter calls the Parallel Postulate is equivalent (in both senses!) to what *Experiencing Geometry* calls the High School Parallel Postulate (HSP), because it is commonly assumed in high school geometry textbooks.

Chapter 10

Final Projects

For your final project, pick a geometric topic that we have not covered in either semester of this course and explore it on your own.

Your project will consist of three parts:

1. A written paper;
2. A ten minute in-class presentation; and
3. A model of some kind related to your topic. Your model can be a physical model or a computer model.

You can look at any topic, as long as I approve it. Many of my suggested topics come from parts of the *Experiencing Geometry* textbook that we have not studied previously. Each project must include some proof(s) of some kind. People working on the same or related topics may work together, but a group of two people will be expected to cover twice as much material as a single person working alone.

Some possible topics:

- Triangle congruence theorems: ASA (Problem 6.5), SSS (9.1), ASS (9.2), SAA (9.3), AAA (9.4)
- Parallel Postulates: 10.2, 10.3, 10.4
- Patterns: 11.5, 11.6
- Spherical Projections and map making: 14.1 14.4. (14.1 + 14.2 together count as one problem, and only for a group also doing 14.3 and/or 14.4)
- Upper half-plane and Poincaré Disk models: 17.1-17.6
- 2-Manifolds (Flat torus, Klein bottle, etc.): (18.1+18.2) (flat), 18.3 (spherical), 18.4 (hyperbolic), 18.5 (Euler Number), 18.6 (2D Shape of Space)
- Dissection Theory: Ch. 12, 13.
- Geometric algebra; Euclid's *Elements* Book II; and Geometric solutions of Quadratic and Cubic Equations (19.1-19.4)

- Trigonometry and Duality: 20.1 (intrinsic circumference of a circle), 20.2 (spherical law of cosines), 20.3 (spherical law of sines), (20.4 + 20.5)(Duality and the laws of sines/cosines) or (20.4 + 20.6) (Duality + Projective plane) (relates to perspective drawing and 18.3)
- Mechanisms 21.1, 21.2 (Four bar linkages); 21.3 (Reuleaux triangles)
- 4D space: 22.1, only if in a group doing other problems from this chapter; 22.2 (3-Spheres), 22.3-22.6
- Polyhedra: 23.1-23.4 (Spherical angles and tetrahedra congruence theorems)
- Demi-regular tessellations, symmetries of tessellations, etc.
- Tessellations of $\mathbf{R}^3$ by polyhedra (This topic is related to chemistry and how crystals are formed.)
- Questions related to astronomy: astrolabe, calendars, sundial, orrery
- Golden Ratio

Appendix A

Comment Sheets

The following sections contain comments that are frequently made on papers in this course. If you get a paper back with a comment that is just a letter and number in a box, look here to find the corresponding comment.

A.1 Hyperbolic Plane Comments

H1. How do you know that radial lines have the symmetries of a straight line? Problem 5.1 suggests that you use the local intrinsic symmetries of the annular strips. Which symmetry of straight lines do the radial lines clearly have on a single annular strip?

H2. Good. What property of straight lines does this relate to?

H3. The easiest way to show that all right angles, at all points on the hyperbolic plane, are equal to one another is to show that all angles that meet Euclid's definition of being right are 90 degrees, and therefore they're all equal.

H4. Can you give a specific counterexample for Euclid's fifth postulate?

H5. Ok—but how do you know that a straight line can be drawn connecting any two points? Can you describe how you would actually do it given two points on a physical hyperbolic plane?

H6. The word “parallel” normally is defined to mean “non-intersecting.”

H7. There are lines that intersect on the hyperbolic plane, as well as lines that don't intersect and don't get close together. The asymptotic lines are very special in that they get arbitrarily close together, but don't intersect.

H8. How do you know that the radial lines can never intersect?

H9. How does this help show that Euclid's Fifth postulate is false?

A.2 Folding a Parallel Line Comments

A1. You need to carefully show that each part of your figure lands on another part of the figure when rotated 180 degrees about the center point, like we did in class. Start from the center and work your way outward.

A2. How do you know that this always creates a perpendicular line? Can you prove that the lines are perpendicular?

A3. The word “parallel,” used without any qualification, is normally understood to mean “non-intersecting.” Is that what you mean here?

A4. How do you know that the figure that you folded produces parallel transports?

A5. Why does this mean that the two lines can’t ever intersect?

A.3 Parallel Postulates Comments

B1. Did you start with the right beginning? Remember, if you’re trying to prove a statement B by assuming statement A , and if B is an *if-then* statement, then you need to assume the *if* part of B is true and demonstrate in that case that the *then* part of B is true. Somewhere in this argument you will have to use the fact that A is true.

B2. If you’re showing that two statements can be used to prove one another, you can’t assume that you’re working in the plane. Statements are simply true or false in the plane. Proofs, on the other hand, take place in an assumed axiom system, and therefore deal with all possible models of the assumed axioms.

B3. What, exactly, are the two different definitions of the word “equivalent” that you are working with? You need to explicitly state them.

B4. You’ve shown there is at most one line through the point parallel to the original line. But how do you know that there are any lines through the point that are parallel to the original line?

B5. What is the sense in which the two postulates are not equivalent?

B6. Why must this line be parallel to your original line?

B7. Good! Why does your proof that EFP and PP can be used to prove one another show that the same conclusions can be drawn from each of them?

Appendix B

From Book I of Euclid's *Elements*

Euclid wrote the thirteen books of the *Elements* around 300 B.C. The first four books are about elementary planar geometry; books five through ten are largely about the theory of ratio and proportion and about what would now be considered number theory; and books eleven through thirteen are about three dimensional geometry.

The excerpt reprinted here is taken from the edition translated and edited by Thomas Heath [Euclid, 1956].

B.1 Definitions

1. A **point** is that which has no part.
2. A **line** is breadthless length.
3. The extremities of a line are points.
4. A **straight** line is a line which lies evenly with the points on itself.
5. A **surface** is that which has length and breadth only.
6. The extremities of a surface are lines.
7. A **plane** surface is a surface which lies evenly with the straight lines on itself.
8. A **plane angle** is the inclination to one another of two lines in a plane which meet one another and do not lie in a straight line.
9. And when the lines containing the angle are straight, the angle is called **rectilineal**.
10. When a straight line set up on a straight line makes the adjacent angles equal to one another, each of the equal angles is **right**, and the straight line standing on the other is called a **perpendicular** to that on which it stands.
11. An **obtuse** angle is an angle greater than a right angle.
12. An **acute** angle is an angle less than a right angle.
13. A **boundary** is that which is an extremity of anything.
14. A **figure** is that which is contained by any boundary or boundaries.

15. A **circle** is a plane figure contained by one line such that all the straight lines falling upon it from one point among those lying within the figure are equal to one another;
16. And the point is called the **centre** of the circle.
17. A **diameter** of the circle is any straight line drawn through the centre and terminated in both directions by the circumference of the circle, and such a straight line also bisects the circle.
18. A **semicircle** is the figure contained by the diameter and the circumference cut off by it. And the centre of the semicircle is the same as that of the circle.
19. **Rectilineal figures** are those which are contained by straight lines, **trilateral** figures being those contained by three, **quadrilateral** those contained by four, and **multilateral** those contained by more than four straight lines.
20. Of trilateral figures, an **equilateral** triangle is that which has its three sides equal, an **isosceles** triangle that which has two of its sides alone equal, and a **scalene** triangle that which has its three sides unequal.
21. Further, of trilateral figures, a **right-angled** triangle is that which has a right angle, an **obtuse-angled** triangle that which has an obtuse angle, and an **acute-angled** triangle that which has its three angles acute.
22. Of quadrilateral figures, a **square** is that which is both equilateral and right-angled; an **oblong** [rectangle] that which is right-angled but not equilateral; a **rhombus** that which is equilateral but not right-angled; and a **rhomboid** [parallelogram] that which has its opposite sides and angles equal to one another but is neither equilateral nor right-angled. And let quadrilaterals other than these be called **trapezia**.
23. **Parallel** straight lines are straight lines which, being in the same plane and being produced indefinitely in both directions, do not meet one another in either direction.

B.2 Postulates

Let the following be postulated:

1. To draw a straight line from any point to any point.
2. To produce a finite straight line continuously in a straight line.
3. To describe a circle with any centre and distance.
4. That all right angles are equal to one another.
5. That, if a straight line falling on two straight lines make the interior angles on the same side less than two right angles, the two straight lines, if produced indefinitely, meet on that side on which are the angles less than the two right angles.

B.3 Common Notions

1. Things which are equal to the same thing are also equal to one another.
2. If equals be added to equals, the wholes are equal.
3. If equals be subtracted from equals, the remainders are equal.
4. Things which coincide with one another are equal to one another.
5. The whole is greater than the part.

Appendix C

Problems from *Experiencing Geometry*

These problems are taken from *Experiencing Geometry: Euclidean and non-Euclidean with History* [Henderson & Tamiņa, 2005].

Problem 1.1: When Do You Call a Line Straight?

- a. How can you check in a practical way if something is straight? How do you construct something straight—lay out fence posts in a straight line, or draw a straight line? Do this without assuming that you have a ruler, for then we will ask, “How can you check that the ruler is straight?”
- b. What symmetries does a straight line have?
- c. What is in common among the different notions of straightness? Can you write a definition of “straight line”?

Problem 2.1: What is Straight on a Sphere?

- a. Imagine yourself to be a bug crawling around on a sphere. (This bug can neither fly nor burrow into the sphere.) The bug’s universe is just the surface; it never leaves it. What is “straight” for this bug? What will the bug see or experience as straight? How can you convince yourself of this? Use the properties of straightness (such as symmetries) that you talked about in Problem 1.1.

Problem 5.1: What is Straight in a Hyperbolic Plane?

- a. On a hyperbolic plane, consider the curves that run radially across each annular strip. Argue that these curves are intrinsically straight. Also, show that any two of them are asymptotic, in the sense that they converge toward each other but do not intersect.
- b. Find other geodesics on your physical hyperbolic surface. Use the properties of straightness (such as symmetries) you talked about in Problems 1.1, 2.1, and 4.1.
- c. What properties do you notice for geodesics on a hyperbolic plane? How are they the same as geodesics on the plane or spheres, and how are they different from geodesics on the plane and spheres?

Problem 7.1: The Area of a Triangle on a Sphere

- The two sides of each interior angle of a triangle Δ on a sphere determine two congruent lunes with lune angle the same as the interior angle. Show how the three pairs of lunes determined by the three interior angles α , β , γ , cover the sphere with some overlap. (What is the overlap?)
- Find a formula for the area of a lune with lune angle θ in terms of θ and the (surface) area of the sphere (of radius ρ), which you can call S_ρ . Use radian measure for angles.
- Find a formula for the area of a triangle on a sphere of radius ρ .

Problem 7.2: Area of Hyperbolic Triangles

- On your hyperbolic plane draw as large a triangle as you can find. Compare your triangle with the large triangles that others have found. What do you notice?
- Show that on the same hyperbolic plane, all 2/3-ideal triangles with the same angle θ are congruent. [A 2/3-ideal triangle is one with two vertices at infinity.]
- Show that the area function A_ρ is an additive function. That is,

$$A_\rho(\alpha + \beta) = A_\rho(\alpha) + A_\rho(\beta).$$

[Here, $A_\rho(\alpha)$ denotes the area of a 2/3-ideal triangle with exterior angle α on a hyperbolic plane of radius ρ .]

- Find a formula for the area of a hyperbolic triangle.

Problem 8.2: Symmetries of Parallel Transported Lines

Consider two lines, r and r' , that are parallel transports of each other along a third line, l . [Two lines r and r' are said to be parallel transports along a transversal l if l cuts r and r' at congruent angles.] Consider now the geometric figure that is formed by the three lines, one of them being a transversal to the other two, and look for the symmetries of that geometric figure.

What can you say about the lines r and r' ? Do they intersect? If so, where? Look at the plane, spheres, and hyperbolic planes.

Problem 8.3: Transversals through a Midpoint

- Prove: If two geodesics r and r' are parallel transports along another geodesic l , then they are also parallel transports along any geodesic passing through the midpoint of the segments between r and r' . Does this hold for the plane, spheres, and hyperbolic planes?
- On a sphere or hyperbolic plane, are these the only lines that will cut r and r' at congruent angles? Why?

- c. Prove: Two geodesics (on the plane, spheres, or hyperbolic planes) are parallel transports of each other if and only if they have a common perpendicular. On spheres and hyperbolic planes the common perpendicular is unique.

Problem 8.4: What is “Parallel”?

...[Consider] these common definitions for parallel lines in the plane:

1. Parallel lines are lines that never intersect;
 2. Parallel lines are lines such that any transversal cuts them at congruent angles; or
 3. Parallel lines are lines that are everywhere equidistant.
- a. Check out for each of these three definitions whether they apply to parallel transported lines on a sphere or on a hyperbolic plane.
- b. Show that there are pairs of geodesics on a hyperbolic plane that do not intersect and yet there are **no** transversals that cut at congruent angles. That is, the geodesics are parallel (in the sense of not intersecting) but not parallel transports of each other.
- c. Show that there are pairs of geodesics on any cone with cone angle greater than 360° that do not intersect and yet there are **no** transversals that cut at congruent angles. That is, the geodesics are parallel (in the sense of not intersecting) but not parallel transports of each other along any straight (in the sense of symmetry) line.

Appendix D

Excerpt from *Gödel, Escher, Bach*

The following excerpt is taken from Chapter 4 of *Gödel, Escher, Bach: an Eternal Golden Braid* by Douglas Hofstadter [Hofstadter, 1999, pp. 88–93]. This book won the Pulitzer Prize for general non-fiction in 1980.

Prior to this excerpt, Hofstadter has been discussing a formal system that he calls the pq-system which allows you to derive certain strings of symbols. An example of a statement provable in the pq-system is

$$- - - p - - q - - - - - .$$

Hofstadter points out that all of the statements in the pq-system will correspond to true statements of arithmetic if we interpret a statement like this as meaning “three plus two equals five”. However, he then points out that they are equally all true if we interpret it as meaning “three equals two subtracted from five” or “three plus two is less than or equal to five”. Thus, there are multiple possible meaningful interpretations of the statements of the formal system.

The excerpt follows.

The History of Euclidean Geometry

Although I have tried to catch you off guard and surprise you a little, this lesson about how to interpret symbols by words may not seem terribly difficult once you have the hang of it. In fact, it is not. And yet it is one of the deepest lessons of all of nineteenth century mathematics! It all begins with Euclid, who, around 300 B.C., compiled and systematized all of what was known about plane and solid geometry in his day. The resulting work, Euclid’s *Elements*, was so solid that it was virtually a bible of geometry for over two thousand years—one of the most enduring works of all time. Why was this so?

The principal reason was that Euclid was the founder of rigor in mathematics. The *Elements* began with very simple concepts, definitions, and so forth, and gradually built up a vast body of results organized in such a way that any given result depended only on foregoing results. Thus, there was a definite plan to the work, an architecture which made it strong and sturdy.

Nevertheless, the architecture was of a different type from that of, say, a skyscraper. . . . In the latter, that it is standing is proof enough that its structural elements are holding it up. But in a book on geometry, when each proposition is claimed to follow logically from earlier propositions, there will be no visible crash if one of the proofs is invalid. The

girders and struts are not physical, but abstract. In fact, in Euclid's *Elements*, the stuff out of which proofs were constructed was human language—that elusive, tricky medium of communication with so many hidden pitfalls. What, then, of the architectural strength of the *Elements*? Is it certain that it is held up by solid structural elements, or could it have structural weaknesses?

Every word which we use has a meaning to us, which guides us in our use of it. The more common the word, the more associations we have with it, and the more deeply rooted is its meaning. Therefore, when someone gives a definition for a common word in the hopes that we will abide by that definition, it is a foregone conclusion that we will not do so but will instead be guided, largely unconsciously, by what our minds find in their associative stores. I mention this because it is the sort of problem which Euclid created in his *Elements*, by attempting to give definitions of ordinary, common words such as “point”, “straight line”, “circle”, and so forth. How can you define something of which everyone already has a clear concept? The only way is if you can make it clear that your word is supposed to be a technical term, and is not to be confused with the everyday word with the same spelling. You have to stress that the connection with the everyday word is only suggestive. Well, Euclid did not do this, because he felt that the points and lines of his *Elements* were indeed *the* points and lines of the real world. So by not making sure that all associations were dispelled, Euclid was inviting readers to let their powers of association run free . . .

This sounds almost anarchic, and is a little unfair to Euclid. He did set down axioms, or postulates, which were supposed to be used in the proofs of propositions. In fact, nothing other than those axioms and postulates was supposed to be used. But this is where he slipped up, for an inevitable consequence of his using ordinary words was that some of the images conjured up by those words crept into the proofs which he created. However, if you read proofs in the *Elements*, do not by any means expect to find glaring “jumps” in the reasoning. On the contrary, they are very subtle, for Euclid was a penetrating thinker, and would not have made any simple-minded errors. Nonetheless, gaps are there, creating slight imperfections in a classic work. But this is not to be complained about. One should merely gain an appreciation for the difference between absolute rigor and relative rigor. In the long run, Euclid's lack of absolute rigor was the cause of some of the most fertile path-breaking in mathematics, over two thousand years after he wrote his work.

Euclid gave five postulates to be used as the “ground story” of the infinite skyscraper of geometry, of which his *Elements* constituted only the first several hundred stories. The first four postulates are rather terse and elegant:

1. A straight line segment can be drawn joining any two points.
2. Any straight line segment can be extended indefinitely in a straight line.
3. Given any straight line segment, a circle can be drawn having the segment as radius and one end point as center.
4. All right angles are congruent.

The fifth, however, did not share their grace:

5. If two lines are drawn which intersect a third in such a way that the sum of the inner angles on one side is less than two right angles, then the two lines inevitably must intersect each other on that side if extended far enough.

Though he never explicitly said so, Euclid considered this postulate to be somehow inferior to the others, since he managed to avoid using it in the proofs of the first twenty-eight

propositions. Thus, the first twenty-eight propositions belong to what might be called “four-postulate geometry”—that part of geometry which can be derived on the basis of the first four postulates of the *Elements*, without the help of the fifth postulate. (It is also often called *absolute geometry*.) Certainly Euclid would have found it far preferable to *prove* this ugly duckling, rather than to have to *assume* it. But he found no proof, and therefore adopted it.

But the disciples of Euclid were no happier about having to assume this fifth postulate. Over the centuries, untold numbers of people gave untold years of their lives in attempting to prove that the fifth postulate was itself part of four-postulate geometry. By 1763, at least twenty-eight different proofs had been published—all erroneous! (They were all criticized in the dissertation of one G. S. Klügel.) All of these erroneous proofs involved a confusion between everyday intuition and strictly formal properties. It is safe to say that today, hardly any of these “proofs” holds any mathematical or historical interest—but there are certain exceptions.

The Many Faces of Noneuclid

Girolamo Saccheri (1667–1733) lived around Bach’s time. He had the ambition to free Euclid of every flaw. Based on some earlier work he had done in logic, he decided to try a novel approach to the proof of the famous fifth: suppose you *assume its opposite*; then work with *that* as your fifth postulate . . . Surely after a while you will create a contradiction. Since no mathematical system can support a contradiction, you will have shown the unsoundness of your own fifth postulate, and therefore the soundness of Euclid’s fifth postulate. We need not go into details here. Suffice it to say that with great skill, Saccheri worked out proposition after proposition of “Saccherian geometry” and eventually became tired of it. At one point, he decided he had reached a proposition which was “repugnant to the nature of the straight line”. That was what he had been hoping for—to his mind, it was the long-sought contradiction. At that point, he published his work under the title *Euclid Freed of Every Flaw*, and then expired.

But in so doing, he robbed himself of much posthumous glory, since he had unwittingly discovered what came later to be known as “hyperbolic geometry”. Fifty years after Saccheri, J. H. Lambert repeated the “near miss”, this time coming even closer, if possible. Finally, forty years after Lambert, and ninety years after Saccheri, *non-Euclidean geometry* was recognized for what it was—an authentic new brand of geometry, a bifurcation in the hitherto single stream of mathematics. In 1823, non-Euclidean geometry was discovered simultaneously, in one of those inexplicable coincidences, by a Hungarian mathematician, János (or Johann) Bolyai, aged twenty-one, and a Russian mathematician, Nikolay Lobachevskiy, aged thirty. And, ironically, in that same year, the great French mathematician Adrien-Marie Legendre came up with what he was sure was a proof Euclid’s fifth postulate, very much along the lines of Saccheri.

Incidentally, Bolyai’s father, Farkas (or Wolfgang) Bolyai, a close friend of the great Gauss, invested much effort in trying to prove Euclid’s fifth postulate. In a letter to his son János, he tried to dissuade him from thinking about such matters:

You must not attempt this approach to parallels. I know this way to its very end. I have traversed this bottomless night, which extinguished all light and joy of my life. I entreat you, leave the science of parallels alone. ... I thought I would sacrifice myself for the sake of the truth. I was ready to become a martyr who would remove the flaw from geometry and return it purified to mankind.

I accomplished monstrous, enormous labors; my creations are far better than those of others and yet I have not achieved complete satisfaction. For here it is true that *si paullum a summo discessit, vergit ad imum*. I turned back when I saw that no man can reach the bottom of this night. I turned back unconsolated, pitying myself and all mankind. ... I have traveled past all reefs of this infernal Dead Sea and have always come back with broken mast and torn sail. The ruin of my disposition and my fall date back to this time. I thoughtlessly risked my life and happiness—*aut Caesar aut nihil*.¹

But later, when convinced his son really “had something”, he urged him to publish it, anticipating correctly the simultaneity which is so frequent in scientific discovery:

When the time is ripe for certain things, these things appear in different places in the manner of violets coming to light in early spring.²

How true this was in the case of non-Euclidean geometry! In Germany, Gauss himself and a few others had more or less independently hit upon non-Euclidean ideas. These included a lawyer, F. K. Schweikart, who in 1818 sent a page describing a new “astral” geometry to Gauss; Schweikart’s nephew, F. A. Taurinus, who did non-Euclidean trigonometry; and F. L. Wachter, a student of Gauss, who died in 1817, aged twenty-five, having found several deep results in non-Euclidean geometry.

The clue to non-Euclidean geometry was “thinking straight” about the propositions which emerge in geometries like Saccheri’s and Lambert’s. The Saccherian propositions are only “repugnant to the nature of the straight line” if you cannot free yourself of preconceived notions of what “straight line” must mean. If, however, you can divest yourself of those preconceived images, and merely let a “straight line” be something which satisfies the new propositions, then you have achieved a radically new viewpoint.

Undefined Terms

This should begin to sound familiar. In particular, it harks back to the pq-system, and its variant, in which the symbols acquired passive meanings by virtue of their roles in theorems. The symbol q is especially interesting, since its “meaning” changed when a new axiom schema was added. In that very same way, one can *let the meanings of “point”, “line”, and so on be determined by the set of theorems (or propositions) in which they occur*. This was the great realization of the discoverers of non-Euclidean geometry. They found different sorts of non-Euclidean geometries by denying Euclid’s fifth postulate in different ways and following out the consequences. Strictly speaking, they (and Saccheri) did not deny the fifth postulate directly, but rather, they denied an equivalent postulate, called the *parallel postulate*, which runs as follows:

Given any straight line, and a point not on it, there exists one, and only one, straight line which passes through that point and never intersects the first line, no matter how far they are extended.

The second straight line is then said to be parallel to the first. If you assert that no such line exists, then you reach *elliptical geometry*; if you assert that at least two such lines exist, you reach *hyperbolic geometry*. Incidentally, the reason that such variations are still called

¹Herbert Meschkowski, *Non-Euclidean Geometry*, pp. 31–2.

²Ibid., p. 33.

“geometries” is that the core element—absolute, or four-postulate, geometry—is embedded in them. It is the presence of this minimal core which makes it sensible to think of them as describing properties of some sort of geometrical space, even if the space is not as intuitive as ordinary space.

Actually, elliptical geometry is easily visualized. All “points”, “lines”, and so forth are to be parts of the surface of an ordinary sphere. Let us write “POINT” when the technical term is meant, and “point” when the everyday sense is desired. Then, we can say that a POINT consists of a pair of diametrically opposed points of the sphere’s surface. A LINE is a great circle on the sphere (a circle which, like the equator, has its center at the center of the sphere). Under these interpretations, the propositions of elliptical geometry, though they contain words like “POINT” and “LINE”, speak of the goings-on on a sphere, not a plane. Notice that two LINES always intersect in exactly two antipodal points of the sphere’s surface—that is, in exactly one single POINT! And just as two LINES determine a POINT, so two POINTS determine a LINE.

By treating words such as “POINT” and “LINE” as if they had only the meaning instilled in them by the propositions in which they occur, we take a step towards complete formalization of geometry. This semiformal version still uses a lot of words in English with their usual meanings (words such as “the”, “if”, “and”, “join”, “have”), although the everyday meaning has been drained out of special words like “POINT” and “LINE”, which are consequently called undefined terms. Undefined terms, like the p and q of the pq -system, do get defined in a sense: implicitly—by the totality of all propositions in which they occur, rather than explicitly, in a definition.

One could maintain that a full definition of the undefined terms resides in the postulates alone, since the propositions which follow from them are implicit in the postulates already. This view would say that the postulates are implicit definitions of all the undefined terms, all of the undefined terms being defined in terms of the others.

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